

LUHANSK, Ukraine

WMO: 345230

Lat: 48.5656N

Lon: 39.2339E

Elev: 203

StdP: 14.59

Time Zone: 2.00 (EUE)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.7	0.5	-10.7	3.1	-4.8	-5.1	4.2	1.2	16.8	24.4	15.7	21.8	3.0	280	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	95.0	69.1	91.4	68.3	87.6	67.2	71.9	86.9	70.4	85.5	69.1	83.5	7.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	101.2	77.1	65.7	95.6	75.7	64.3	90.8	74.5	35.7	86.2	34.4	85.4	33.3	83.9	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	13.9	12.1	DB	-11.3	98.5	7.5	4.1	-16.7	101.5	-21.1	103.9	-25.3	106.2	-30.7	109.2	(4)
(5)			WB	-11.7	75.8	7.4	2.6	-17.0	77.7	-21.4	79.2	-25.5	80.6	-30.9	82.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	24.7	25.9	35.8	50.4	61.6	69.0	73.6	70.9	60.1	48.5	37.2	27.8
	DBStd	19.50	11.81	11.92	8.19	7.30	7.98	6.32	5.85	6.64	7.43	8.50	9.24	11.03
	HDD50	3216	784	676	443	82	10	0	0	0	9	132	393	687
	HDD65	6586	1249	1096	904	439	162	30	4	19	181	515	835	1152
	CDD50	2819	0	0	4	93	368	570	731	648	313	84	8	0
	CDD65	717	0	0	0	1	56	150	270	203	34	3	0	0
	CDH74	8270	0	0	0	32	731	1613	2996	2454	414	30	0	0
	CDH80	3418	0	0	0	4	218	616	1359	1113	104	4	0	0
Wind	WSAvg	5.1	5.5	6.1	6.6	5.8	4.5	4.4	4.1	4.2	4.4	4.5	5.5	5.8
Precipitation	PrecAvg	18.7	1.3	1.1	1.1	1.6	1.8	2.4	1.9	1.7	1.4	1.3	1.7	1.7
	PrecMax	27.3	3.6	3.4	2.8	5.5	3.5	6.5	5.7	4.1	4.8	3.7	3.5	4.0
	PrecMin	11.1	0.0	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.1	0.2	0.2
	PrecStd	3.7	0.9	0.8	0.6	1.0	0.9	1.5	1.5	1.2	1.1	0.9	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.0	51.2	62.9	78.4	90.6	94.5	99.7	101.2	87.7	78.6	63.1	51.8
		MCWB	45.0	45.9	50.3	58.6	66.2	69.4	69.7	70.0	63.5	62.6	54.4	48.8
	2%	DB	43.8	45.0	57.5	72.7	85.5	90.7	95.4	95.1	82.8	71.0	57.3	47.2
		MCWB	41.5	41.3	46.4	55.9	63.8	67.7	69.1	69.0	63.3	58.3	51.9	44.8
	5%	DB	41.0	41.3	52.2	68.3	81.8	86.9	91.5	90.7	78.8	65.8	52.9	43.4
		MCWB	39.1	39.0	44.5	53.6	62.5	66.5	68.6	68.3	62.1	57.0	49.2	41.6
	10%	DB	38.2	39.0	47.8	64.6	77.8	83.1	87.7	86.0	74.8	61.5	49.4	40.6
		MCWB	36.4	36.8	42.3	52.0	61.3	65.4	68.2	66.6	60.2	55.2	46.5	38.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.1	47.1	51.8	60.7	69.0	72.9	74.7	73.5	68.5	64.0	55.6	49.2
		MCDB	47.1	50.6	61.1	76.2	84.9	87.1	89.6	90.9	80.3	73.7	60.3	51.1
	2%	WB	41.4	41.7	48.0	57.6	66.3	70.3	72.3	71.3	65.9	60.3	52.2	44.8
		MCDB	43.1	43.9	54.4	69.0	80.2	84.0	86.4	88.1	76.6	67.7	56.6	47.1
	5%	WB	39.2	39.1	45.1	55.5	64.5	68.8	70.7	69.7	64.1	58.1	49.6	41.8
		MCDB	40.9	41.2	50.3	64.7	77.1	82.3	85.2	86.5	74.7	64.2	52.5	43.4
	10%	WB	36.4	37.1	42.6	53.6	62.6	67.1	69.4	68.0	62.0	55.7	46.7	39.0
		MCDB	37.9	38.8	47.6	62.0	75.0	79.6	84.0	82.4	71.1	60.7	49.3	40.4
Mean Daily Temperature Range	5% DB	MDBR	9.9	11.6	13.9	20.0	22.5	21.5	22.2	24.4	22.2	16.6	11.3	9.3
		MCDBR	8.4	11.6	22.0	27.8	27.9	27.2	28.2	31.4	28.4	23.9	17.4	10.8
		MCWBR	7.3	9.0	13.1	13.8	11.2	9.9	9.1	11.1	12.5	12.9	12.3	9.0
	5% WB	MCDBR	8.2	10.9	19.0	22.6	25.0	23.3	23.7	26.1	23.0	20.0	15.5	10.6
		MCWBR	7.3	9.0	12.4	11.8	11.2	9.8	9.2	10.8	11.6	11.7	11.7	9.0
Clear-Sky Solar Irradiance	taub		0.333	0.365	0.397	0.459	0.422	0.419	0.445	0.446	0.415	0.381	0.354	0.337
	taud		2.173	2.091	2.137	2.048	2.243	2.286	2.225	2.203	2.251	2.305	2.288	2.216
	Ebn at Noon		220	239	254	250	267	268	259	252	247	235	213	201
	Edh at Noon		26	36	41	49	42	41	43	42	36	29	24	23
All-Sky Solar Radiation	RadAvg		315	615	948	1372	1840	1988	1943	1772	1232	699	325	200
	RadStd		71	128	151	148	161	166	124	123	123	106	64	29

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.22	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+295	+166
(47) Regional (7 neighbors)		+1.22	N/A	N/A	N/A	N/A	N/A	N/A	-311	+211	+133

Nomenclature: See separate page